No.	Status	Date
QSCP-NDT-001 Rev 3	Issued	2024-01-15

Site NDE Procedure - Ultrasonic Thickness Measurement, ATU-100

Scope and Application

This procedure establishes the requirements for ultrasonic thickness measurement within the ATU-100 to ensure compliance with QSCP-CIS-001 Rev 1. It applies to all piping circuits, including Class 1 sour gas and regenerator overhead lines, Class 2 amine and treated gas lines, and Class 3 utility systems. The methodology focuses on standardized data acquisition to support the risk-based inspection intervals defined by the corporate integrity standard.

This document supersedes QSCP-NDT-001 Rev 2 and incorporates findings from QSCP-RBI-ATU100-2023 regarding data quality and CML-id consistency. All NDE personnel must adhere to the digital export and naming conventions specified herein to eliminate the mixed date formats and identifier variants observed in previous inspection cycles. The procedure covers both carbon steel and A312 TP316 stainless steel components, utilizing specific velocity settings for each material.

Responsibilities

Jennifer Walsh, as the Inspection Coordinator, is responsible for the technical oversight of NDE contractors and ensuring that all personnel hold valid CAN/CGSB-48.9712 or EN ISO 9712 certifications. The Inspection Coordinator shall verify that all technicians have completed the mandatory 3-year site orientation before performing work on the unit. Furthermore, Jennifer Walsh must audit the digital data exports to ensure alignment with the corporate CML-id format.

Robert Hendricks, acting as the Unit Engineer, is responsible for reviewing inspection results that fall below the established alert levels. Robert Hendricks shall authorize any necessary repairs or fitness-for-service assessments triggered by readings that approach the retirement thickness. The Unit Engineer also maintains final approval over the calibration-interval table and any deviations from the standard RBI intervals.

Calibration and Measurement Requirements

Technicians shall utilize the calibration-interval table to maintain instrument accuracy throughout the work shift. For all readings that fall below the designated alert level, a photo of the gauge display must be captured and included in the digital export. This visual verification is mandatory for all Class 1 and Class 2 piping to support the findings of QSCP-RBI-ATU100-2023 and ensure high-confidence data for corrosion rate calculations.

Measurements on insulated lines within the CUI range must be coordinated with the CUI program requirements, utilizing profile RT or insulation windows every 5 years. All digital data must be exported in the format required by QSCP-CIS-001 Rev 1 to facilitate the calculation of long-term and short-term corrosion rates. The larger of these two rates will be used to determine the remaining life and the subsequent inspection due date.

Data Evaluation and Reporting

The retirement thickness for any component is defined as the line t-min. Any reading recorded below this value must be immediately raised as an NCR and reported to the Inspection Coordinator. Remaining life is calculated by subtracting t-min from the actual thickness and dividing by the governing corrosion rate. The next inspection interval shall not exceed half of the remaining life, subject to the class maxima of 5 or 10 years.

Final reports must be submitted digitally, ensuring that all CML-id entries match the master registry to prevent the errors identified in QSCP-NDT-001 Rev 2. The integration of PAUT for corrosion mapping, where required by the corporate standard, shall follow the same digital reporting rules. All records are to be archived within the asset integrity management system to maintain a continuous history for the ATU-100.

Sign-off	Name	Function	Date	Signature
Prepared	Jennifer Walsh	Inspection Coordinator, API 570 inspector	2024-01-15	_____
Approved	Robert Hendricks	Unit Engineer ATU-100, P.Eng.	2024-01-15	_____